Fundamentals of Data Ethics and Legal Obligations

Week 3

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Course; Certificate IV IT (Data Science and AI)

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Learning Objectives

- Define fundamental data ethics principles and industry standards in AI.
- Identify relevant data protection laws and organizational responsibilities.
- Analyze practical scenarios using open datasets with an ethical and legal perspective.

Session Agenda

- Why data ethics is critical in AI and data science.
- Key ethical frameworks and standards for responsible data use.
- Data protection laws; GDPR; Australian Privacy Act.
- Organisational codes of conduct and accountability.
- Industry example; class activity.
- Assessment links and review.

Why Data Ethics Matters

- AI relies on large, open datasets; potential for harm if mismanaged.
- Ethical and legal compliance protect individuals and organizations.
- Responsible practice is foundational for modern AI and data careers.

Data Ethics Principles

- Transparency; make data practices clear and understandable.
- Fairness; avoid bias and discrimination in dataset handling.
- Privacy; protect personal and sensitive information.
- Accountability; ensure oversight and responsibility for data decisions.

Key Frameworks and Standards

- ACM Code of Ethics; guides IT professionals.
- European GDPR; comprehensive data rights standard.
- Australian Privacy Act; governs personal data use in Australia.
- Fair Information Practice Principles; global data management best practices.
- Responsible AI principles; used by leading tech firms and open data consortia.

Data Protection Laws

- General Data Protection Regulation (GDPR); applies to data about EU citizens; impacts international dataset sharing.
- Australian Privacy Act; sets rules for organizations handling personal data in Australia.
- Laws affect preprocessing, sharing, and usage of large AI datasets.

Organisational Responsibilities

- Codes of conduct; outline employee obligations in data handling.
- Policies and governance; ensure compliance within AI/data teams.
- Training and systems; monitor, document, and report data practices.

Industry Example

Responsible Dataset Use in AI

- An Australian tech company wants to use LAION-5B for training LLMs.
- Must apply privacy filters; document sources; comply with GDPR and Privacy Act.
- Team reviews dataset for bias and fairness issues before project starts.

In-Class Activity

- Small group discussion; review a real-world AI dataset scenario.
- Identify; potential ethical and legal risks; propose mitigation steps.
- Share findings; discuss how these issues relate to your future roles.

Preparing for Assessment

- Connecting theory to practice; research report will assess your ability to analyze a large open dataset for ethics and compliance.
- Review points; highlight what to look for in documentation, consent, and risk factors.
- Knowledge-based test; key terms and concepts from this session will appear.

Summary and Key Takeaways

- Data ethics and legal compliance are non-negotiable in AI and data industries.
- Understand and apply transparency, fairness, privacy, and accountability.
- Know major laws and codes that guide responsible data work.

Next Steps and Course Progression

- Next week; Essential Python Tools for Data Science.
- Use this week’s knowledge as foundation for assessing dataset quality (Weeks 5-6).
- Review notes; revisit Week 1-2 for open data context; prepare questions for next session.